



CS 260 – Foundations of Data Science

Lecture 8 – Naïve Bayes

09/24/2026

Adam Poliak



Announcements

HW4 due Tuesday night

Midterm 1 – October 6th

Office Hours:

Tuesdays: 4pm-4:30pm

Thursdays 2-3:30pm, Park 200D

Today just until 12:00



Outline

Probability

Intro to Bayesian Models

Naïve Bayes Algorithm

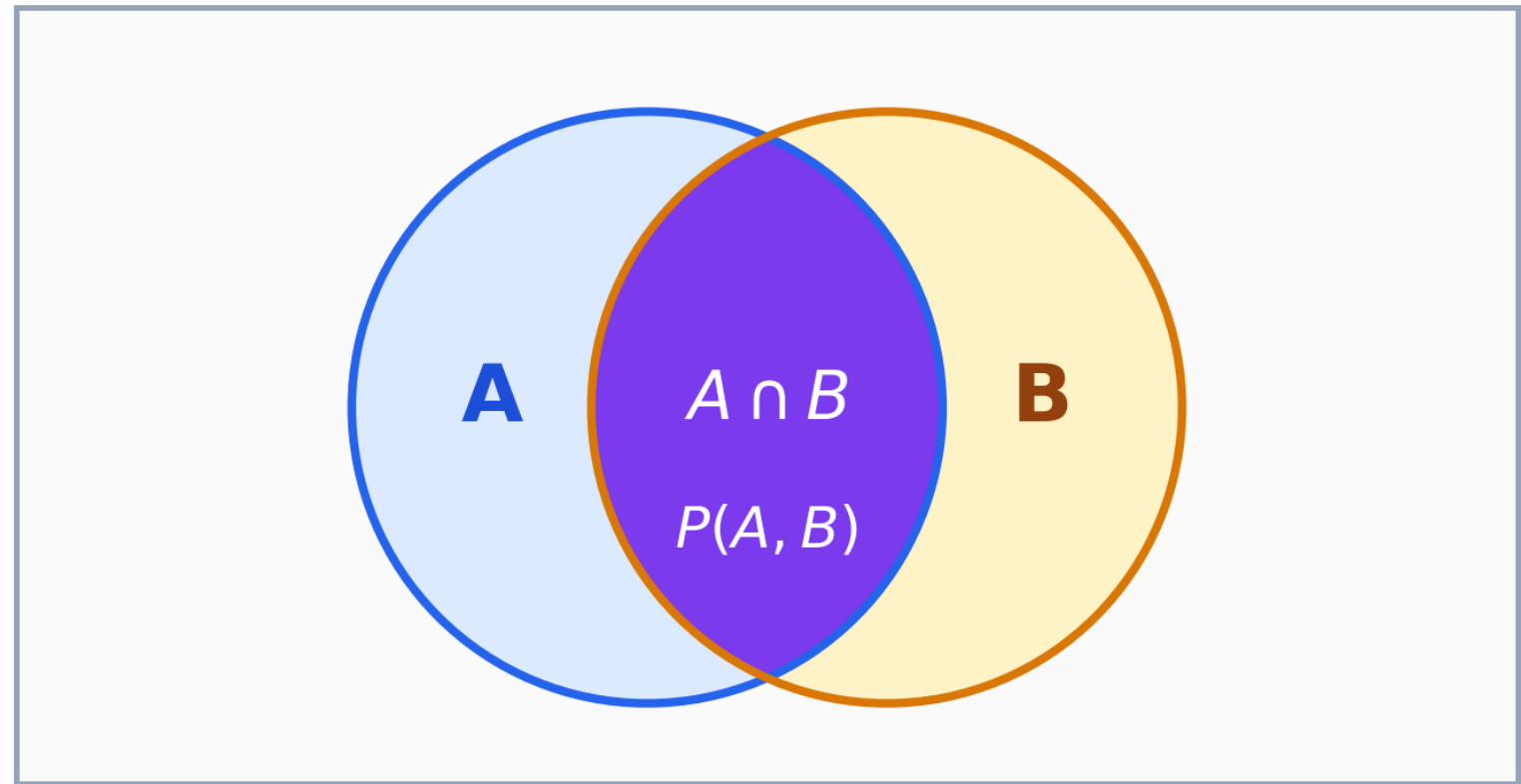


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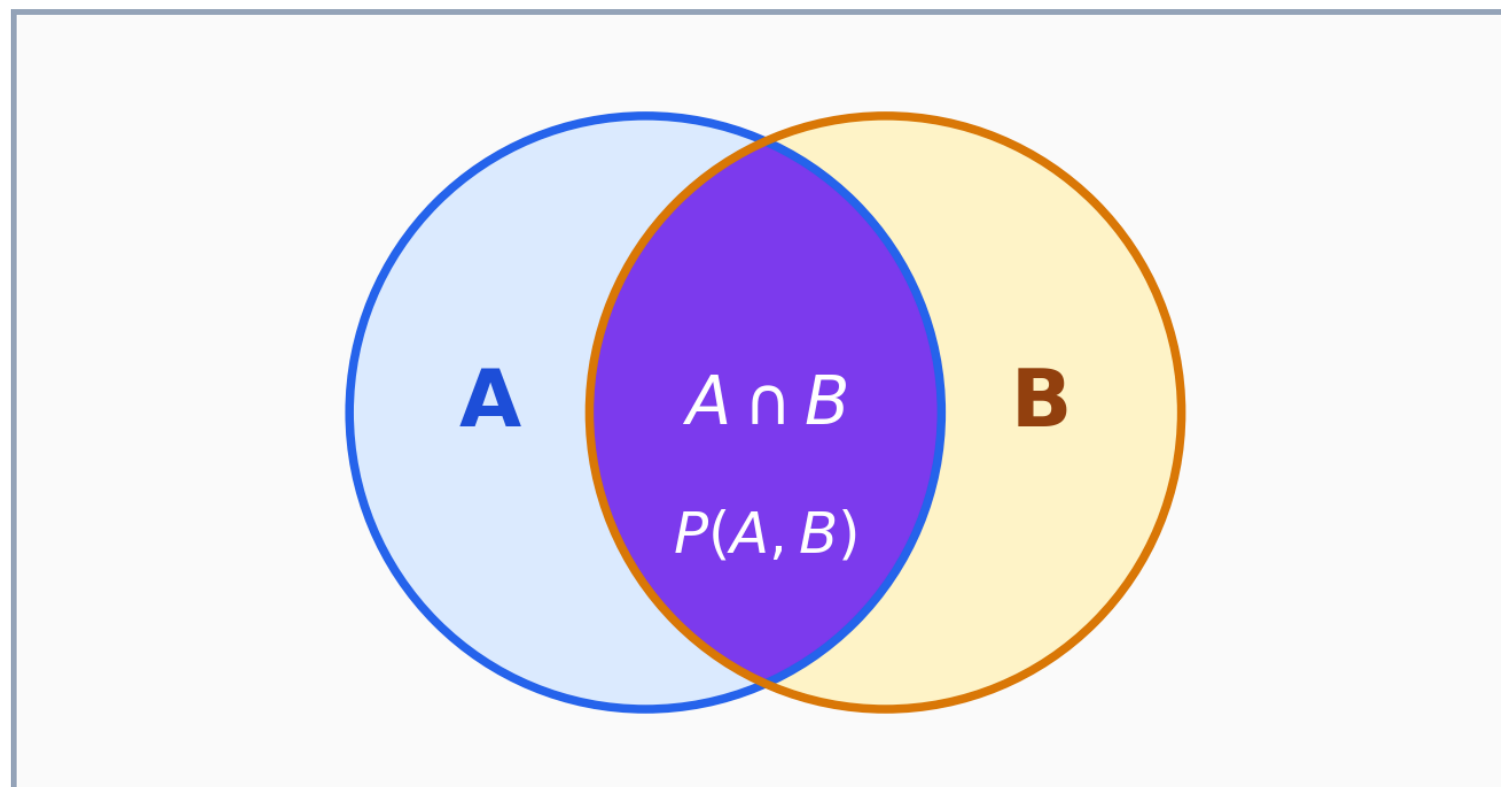
Bayes Rule

$$P(A, B) =$$



Bayes Rule

$$\begin{aligned}P(A, B) &= \\&= P(A)P(A|B) \\&= P(B)P(B|A)\end{aligned}$$



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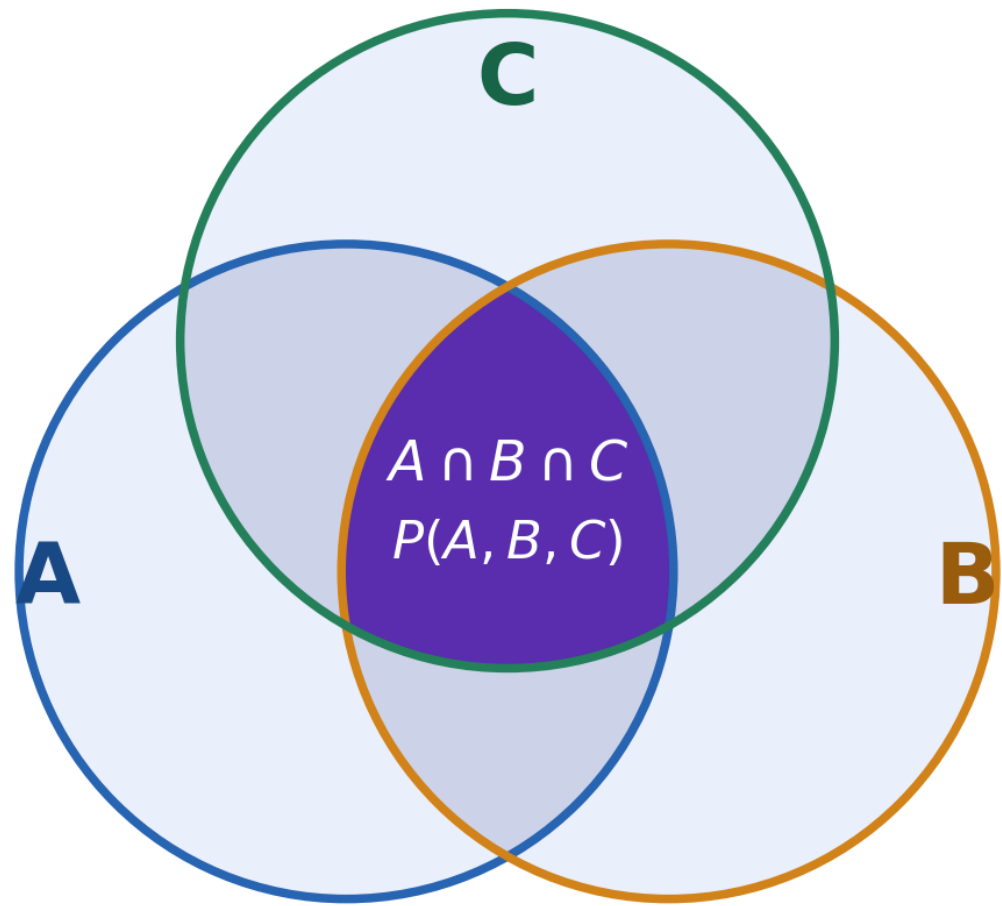
Hence:

$$P(A|B) = \frac{P(A|B)P(A)}{P(B)}$$



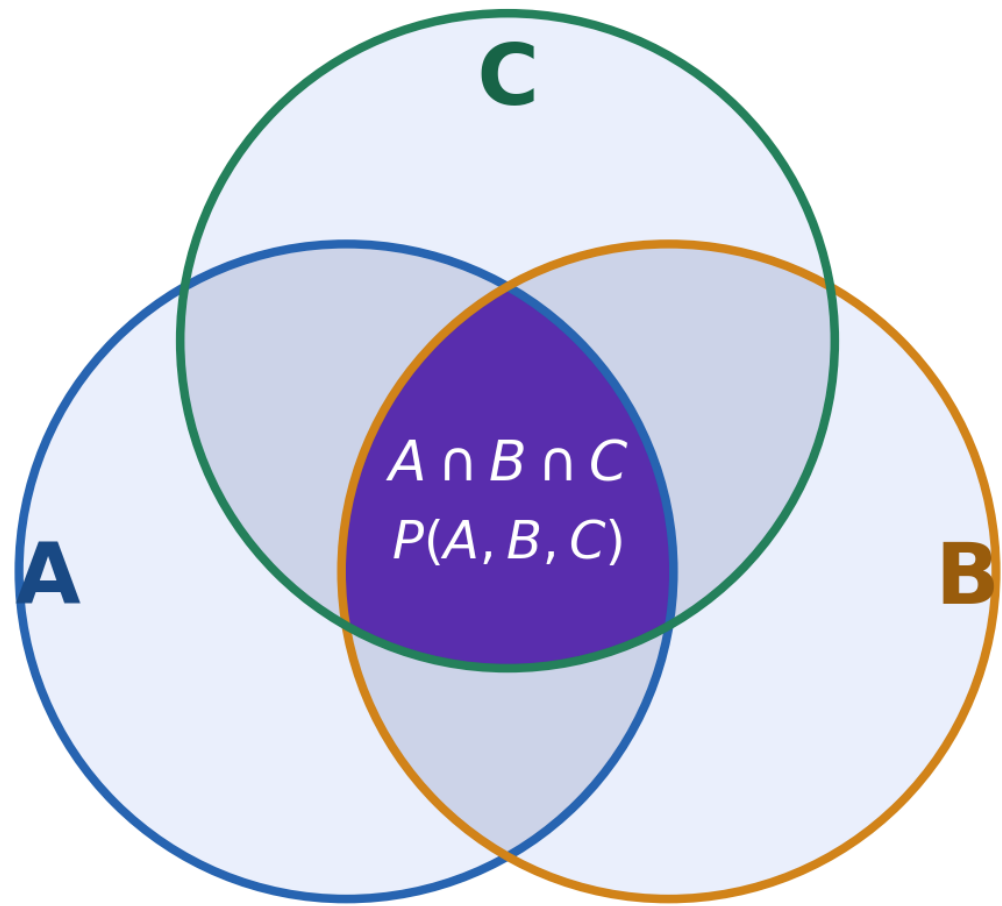
Multiple Variables

$$P(A, B, C) =$$



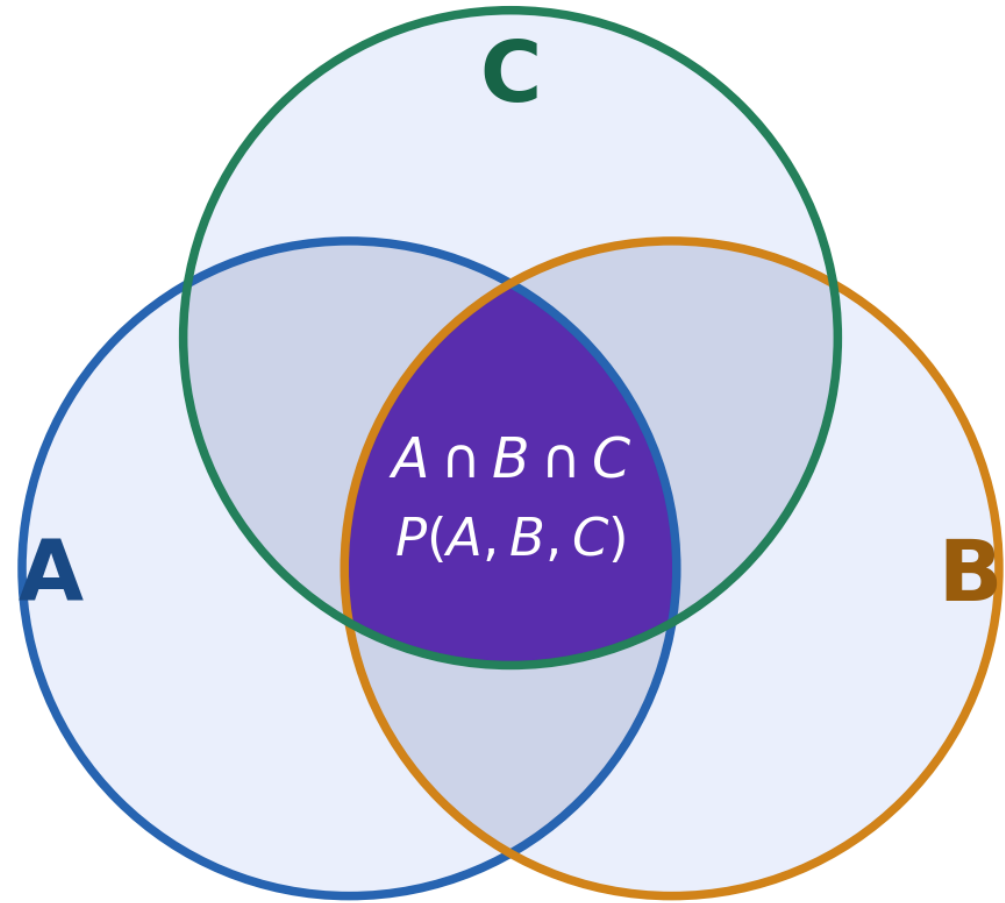
Multiple Variables

$$P(A, B, C) = P(A)P(B|A)P(C|B, A)$$



Independence

If I learn that A happened,
does that change how likely B
is?



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If no, then $P(B|A) = P(B)$



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$$\begin{aligned} P(A, B) &= P(B|A)P(A) \\ &= P(B)P(A) \end{aligned}$$

$P(A, B) = P(A)P(B)$ is a *special case* of
the joint probability rule



Independence

If A , B , and C are all independent, what is $P(A, B, C)$?

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$$\begin{aligned} P(A, B, C) &= P(A)P(B|A)P(C|B, A) \\ &= P(A)P(B)P(C) \end{aligned}$$



Handout



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Components of a Bayes Model

Identify the evidence, prior, posterior, and likelihood in the equation below

$$p(y = k|\mathbf{x}) = \frac{p(y = k)p(\mathbf{x}|y = k)}{p(\mathbf{x})}$$

Components of a Bayes Model

Identify the **evidence** prior, posterior, and likelihood in the equation below

$$p(y = k | \mathbf{x}) = \frac{p(y = k)p(\mathbf{x} | y = k)}{p(\mathbf{x})}$$

Evidence: this is the data (features) we observe, which we think will help us predict the outcome we're interested in

Components of a Bayes Model

Identify the evidence, prior, posterior, and likelihood in the equation below

$$p(y = k|\mathbf{x}) = \frac{p(y = k)p(\mathbf{x}|y = k)}{p(\mathbf{x})}$$

Prior: without seeing any evidence (data), what is our prior believe about each outcome (intuition: what is the outcome in the population as a whole?)



Components of a Bayes Model

Identify the evidence, prior, posterior, and likelihood in the equation below

$$p(y = k | \mathbf{x}) = \frac{p(y = k)p(\mathbf{x} | y = k)}{p(\mathbf{x})}$$

Posterior: this is the quantity we are actually interested in. **Given** the evidence, what is the probability of the outcome?



Components of a Bayes Model

Identify the evidence, prior, posterior, and likelihood in the equation below

$$p(y = k|\mathbf{x}) = \frac{p(y = k)p(\mathbf{x}|y = k)}{p(\mathbf{x})}$$

Likelihood: given an outcome, what is the probability of observing this set of features?



Examples

- Computing the probability an email message is **spam**, given the **words** of the email
- Another example: what is the probability of **Trisomy 21** (Down Syndrome), given the **amount of sequencing of each chromosome?**



Bayesian Model for Trisomy 21 (T_{21})

Input data are read counts for each chromosome (1,2,...,n):

$$q_1, q_2, \dots, q_n = \vec{q}$$



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Goal:

$$\mathbb{P}(T_{21} | \vec{q}) = \frac{1}{2}$$



Bayesian Model for Trisomy 21 (T_{21})

Input data are read counts for each chromosome (1,2,...,n):

$$q_1, q_2, \dots, q_n = \vec{q}$$

Goal:

$$\begin{aligned}\mathbb{P}(T_{21} | \vec{q}) &= \frac{\mathbb{P}(\vec{q} | T_{21}) \cdot \mathbb{P}(T_{21})}{\mathbb{P}(\vec{q})} \\ &= \frac{\mathbb{P}(\vec{q} | T_{21}) \cdot \mathbb{P}(T_{21})}{\mathbb{P}(\vec{q} | T_{21}) \cdot \mathbb{P}(T_{21}) + \mathbb{P}(\vec{q} | T_{21}^C) \cdot \mathbb{P}(T_{21}^C)}\end{aligned}$$



Bayesian Model for Trisomy 21 (T_{21})

Input data are read counts for each chromosome (1,2,...,n):

$$q_1, q_2, \dots, q_n = \vec{q}$$

Goal:

$$\mathbb{P}(T_{21} | \vec{q}) = \frac{\mathbb{P}(\vec{q} | T_{21}) \cdot \mathbb{P}(T_{21})}{\mathbb{P}(\vec{q})}$$

Prior probability of T_{21}

$$= \frac{\mathbb{P}(\vec{q} | T_{21}) \cdot \mathbb{P}(T_{21})}{\mathbb{P}(\vec{q} | T_{21}) \cdot \mathbb{P}(T_{21}) + \mathbb{P}(\vec{q} | T_{21}^C) \cdot \mathbb{P}(T_{21}^C)}$$

Prior:

$P(T_{21})$

Maternal Age	Trisomy 21	All Trisomies
20	1 in 1,667	1 in 526
21	1 in 1,429	1 in 526
22	1 in 1,429	1 in 500
23	1 in 1,429	1 in 500
24	1 in 1,250	1 in 476
25	1 in 1,250	1 in 476
26	1 in 1,176	1 in 476
27	1 in 1,111	1 in 455
28	1 in 1,053	1 in 435
29	1 in 1,000	1 in 417
30	1 in 952	1 in 384
31	1 in 909	1 in 384
32	1 in 769	1 in 323
33	1 in 625	1 in 286
34	1 in 500	1 in 238
35	1 in 385	1 in 192
36	1 in 294	1 in 156
37	1 in 227	1 in 127
38	1 in 175	1 in 102
39	1 in 137	1 in 83
40	1 in 106	1 in 66
41	1 in 82	1 in 53
42	1 in 64	1 in 42
43	1 in 50	1 in 33
44	1 in 38	1 in 26
45	1 in 30	1 in 21
46	1 in 23	1 in 16
47	1 in 18	1 in 13
48	1 in 14	1 in 10
49	1 in 11	1 in 8



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Real-world example of Naïve Bayes

“A Comparison of Event Models for Naive Bayes Text Classification” (6000+ citations!)

<http://www.kamalnigam.com/papers/multinomial-aaaiws98.pdf>

Goal: text classification (classify documents into topics based on the words as features)

95 topics (i.e., $K=95$)



Naïve Bayes

- Single example: $\vec{x} = [x_1, x_2, \dots, x_p]^T$
- Multi-class label: $y \in \{1, 2, \dots, K\}$
- Goal: Classification $\hat{y} = \operatorname{argmax}_{k=1, \dots, K} p(y = k | \vec{x})$

Bayesian Model

$$p(y = k | \vec{x}) = \frac{p(y = k)p(\vec{x} | y = k)}{p(\vec{x})}$$

can
ignore



Naïve Bayes

$$p(\vec{x}|y = k) = p(\underbrace{x_1}_A, \underbrace{x_2, x_3, \dots, x_p}_B | y = k)$$

$P(A,B)=P(B)P(A|B)$

$$= p(\underbrace{x_2, x_3, \dots, x_p}_{C,D} | y = k) p(\underbrace{x_1}_A | \underbrace{x_2, \dots, x_p}_B, y = k)$$

$$= p(\underbrace{x_3, \dots, x_p}_{D} | y = k) p(\underbrace{x_2}_{C} | \underbrace{x_3, \dots, x_p}_D, y = k) p(x_1 | x_2, \dots, x_p, y = k)$$



Naïve Bayes assumption

Conditional Independence: “feature j is independent from all other features given label k ”

$$p(x_1, x_2 | y) = p(x_1 | y) p(x_2 | x_1, y) \leftarrow \text{Bayes rule}$$

x_1 = 4 legs

x_2 = fur

y = cat

assume $p(x_2 | x_1, y) = p(x_2 | y)$

$$\Rightarrow p(x_1, x_2 | y) = p(x_1 | y) p(x_2 | y)$$



Naïve Bayes

$$\begin{aligned} p(\vec{x}|y = k) &= p(x_p|y = k)p(x_{p-1}|y = k) \dots p(x_2|y = k) p(x_1|y = k) \\ &= \prod_{j=1}^p p(x_j|y = k) \end{aligned}$$

Naïve Bayes Model

$$p(y = k|\vec{x}) \propto p(y = k) \prod_{j=1}^p p(x_j|y = k)$$

proportional to



Obtaining $p(y = k)$ & $p(x_j | y = k)$

Estimate based on training data

- θ_k = estimate for $p(y = k)$
- $\theta_{k,j,v}$ = estimate for $p(x_j = v | y = k)$

feature j

value v

class k

Let N_k = # of examples with label k, we could

define $\theta_k = \frac{N_k}{n}$

What happens if $N_k = 0$?



Naive Bayes Example

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer

Test: powerful very fun

$$p(+) = ?$$

$$p(-) = ?$$



Naive Bayes Example

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Laplacian Smoothing

$$\hat{P}(w_i|c) = \frac{\text{count}(w_i, c) + 1}{\sum_{w \in V} (\text{count}(w, c) + 1)} = \frac{\text{count}(w_i, c) + 1}{(\sum_{w \in V} \text{count}(w, c)) + |V|}$$



Naive Bayes Example

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